Summary in Graph

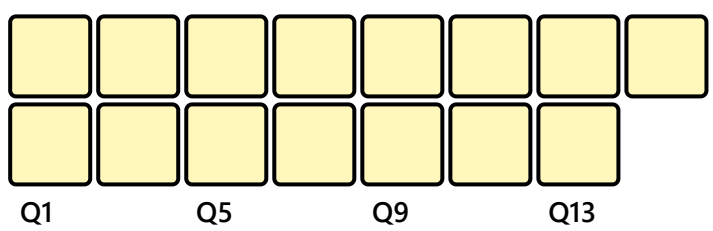
Exam Summary (GO Classes GATE CS | Theory of Computation | Finite Automata and Regular Expression | Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (42) out of 419 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Which of the following four languages is different from the other three?

A. $\emptyset$
B. $\emptyset^*$
C. $\{\lambda\}$

D. $\{\lambda\}^*$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Which of the following four languages is different from the other three?

A. $(\{a, b\})^*$
B. $(\{b, a, a\})^*$
C. $(\{a\}^* \cup \{b\}^*)$
D. $(\{a\} \cup \{b\})^*$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Is there for all non-deterministic finite automata a deterministic finite automaton that recognizes the same language?

A. Yes, you can construct such an automaton using the power set construction.
B. Yes, because you are not required to have non-determinism in a nondeterministic automaton.
C. No, because in a non-deterministic automaton λ transitions are allowed.
D. No, because non-deterministic automata are more efficient than deterministic automata.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Is the complement of a regular language always regular?

A. Yes, because the regular languages are exactly the languages that can be recognized with a DFA, and it is easy to construct a DFA for the complement.
B. Yes, because the complement of the complement is the language itself, so no information is lost by complementation.
C. No, because the complement symbol $\overline{\mathrm{L}}$ is not allowed in a regular expression.
D. No, because if the words in a language are regular, all words that are not regular will be in the complement.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Are there languages for which every deterministic finite automaton has a sink (a non-final state for which each transition will loop back to that state)?

A. Yes, when there is a word w for which any extension ww' will not be in the language.
B. Yes, because each state needs to have a transition for each symbol.
C. No, because you can always split a sink in multiple states.
D. No, because you can always use a non-deterministic automaton.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Are there languages for which every non-deterministic finite automaton does *not* have a sink (a non-final state for which each transition will loop back to that state)?

A. Yes, in the case that each word w has an extension ww' that is in the language, there cannot be a sink.
B. Yes, in non-deterministic automata there never are sinks.
C. No, one can always add a sink to an automaton without changing the language that is accepted.
D. No, there are languages for which each non-deterministic finite automaton has a sink.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let M be a DFA. Say strings x and y in Σ^* are *indistinguishable*, written $x \equiv y$ if and only if

$$\delta^* (q_0, x) = \delta^* (q_0, y)$$

Which of the following is/are necessarily true?

A. Every pair of strings in $\mathrm{L}(\mathrm{M})$ are indistinguishable.
B. Two different strings in $\mathrm{L}(\mathrm{M})$ are never indistinguishable.
C. Indistinguishable relation on strings in Σ^* is an equivalence relation.
D. Indistinguishable relation on strings in Σ^* is a partial order relation.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let M be a DFA. Say strings x and y in Σ^* are *indistinguishable*, written $x \equiv y$ if and only if

$$\delta^* (q_0, x) = \delta^* (q_0, y)$$

Which of the following is/are necessarily true?

A. If strings x, y are indistinguishable then xa, ya are also indistinguishable, for every alphabet symbol a.
B. If strings xa, ya are indistinguishable then x, y are also indistinguishable, where a is an alphabet symbol.
C. If strings x, y are indistinguishable then xw, yw are also indistinguishable, for all strings w.
D. If strings xw, yw are indistinguishable then x, y are also indistinguishable, where w is a string.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Give a deterministic finite automaton with the least number of states for the language: $\mathrm{L}_8 := \{w \in \{a, b\}^* \mid$ the number of a's in w is even iff the number of b's is even $\}$ (The abbreviation 'iff' stands for 'if and only if') What is the number of states?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Give a deterministic finite automaton with the least number of states for the language: $\mathrm{L}_8 := \{w \in \{a,b\}^* \mid$ the number of a's in w is odd iff the number of b's is odd $\}$ (The abbreviation 'iff' stands for 'if and only if') What is the number of states?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

If for a language L is given that $\mathrm{L}^* = \mathrm{L}$, what does not necessarily follow?

A. $\mathrm{L}\,\mathrm{L} = \mathrm{L}$
B. $\lambda \in \mathrm{L}$
C. L is infinite
D. L is non-empty

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

If for a language L is given that $\mathrm{L}^{\mathrm{R}} = \mathrm{L}$, what does necessarily follow?

A. if $w \in \mathrm{L}^*$ then also $w^{\mathrm{R}} \in \mathrm{L}^*$
B. there is a $w \in \mathrm{L}$ for which also $w^{\mathrm{R}} \in \mathrm{L}$
C. $\lambda \in \mathrm{L}$, because $\lambda^{\mathrm{R}} = \lambda$
D. there is a $w \in \mathrm{L}$ with $w^{\mathrm{R}} = w$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let be given a language L. Which of the following languages is not necessarily equal to the others?

A. $\mathrm{L}\,\mathrm{L}^*$
B. $\mathrm{L}\,\mathrm{L}^* \cup \{\lambda\}$
C. $\mathrm{L}^*\mathrm{L}^*$
D. L^*

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Which of the following regular expressions describes a language different from the others?

A. $(a \cup b)^*$
B. $(a^*b^*)^*$

C. $(b^*a^*)^*$
D. $(a^* \cup b^*)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Which of the following regular expressions describes a language different from the others?

A. a^*b^*
B. $(\lambda \cup aa)^*(\lambda \cup bb)^*$
C. $(a \cup aa)^*(b \cup bb)^*$
D. $(\lambda \cup a)^*(\lambda \cup b)^*$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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